
NAMMIIRO ZAHARAH BIDIN

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PROFILE

Passionate fresh graduate software engineer and researcher with a focus on AI , Data , cloud engineering ,and Cybersecurity with AI. Worked on many software projects and also co-founded an AI-driven maternal health platform and led the development of innovative software solutions while on my first professional role in a start up (Syntrix tech Ltd a sister company to Value Addition Microfinance), driving innovation in Uganda. Committed to developing inclusive software solutions that enhance and impact lives through technology. Very Eager to learn and grow in an environment that uses innovation to drive change.

SKILLS

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| <ul style="list-style-type: none">• Application development• Front-end technology• Back-end technology• Software architecture• Software testing• Software Security• Software Infrastructure | <ul style="list-style-type: none">• DevOps• Cloud Architecture(GCP)• Machine learning and AI• Data Science• IoT Technology• Project Management• Collaborative development• Leadership |
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EXPERIENCE

VALUE ADDITION MICROFINANCE

Kampala, Uganda

Software Engineer

June/2025 to May

2026

As a Software Engineer at Value Addition Microfinance (Syntrix Tech Ltd as their sister company), a venture capital firm specializing in technology projects, I oversaw the end-to-end technical architecture and deployment of the firm's most innovative systems. My primary responsibilities involved integrating AI into our solutions while managing the cloud infrastructure for all services via GCP.

Evidence of my architectural leadership includes the successful design of the infrastructure for **Coursemate** (available on the [Google Play Store](#)), a production-level Retrieval-Augmented Generation (RAG) system that grounds LLMs in institutional data to eliminate hallucinations, and the ongoing structural development of the **Nestly** housing application to integrate emerging autonomous technologies. Beyond core development, I lead technical ideation meetings to bridge expertise gaps and drive innovation.

CLUB TANGAZA

Kampala, Uganda

Remote Code Instructor (part time)

01/2023 to 05/2026

- I coordinated and facilitated coding lessons for students in Uganda, the UK, and the USA, enhancing my coding expertise and fostering intercultural exchanges.
- I designed and developed tailored coding curricula for beginners and intermediate learners, focusing on practical programming skills and problem-solving.
- Utilizing online teaching platforms and tools to deliver interactive and engaging coding sessions, ensuring accessibility for students across multiple time zones.

NUFLO
South Africa

Volunteer(Software Engineer)

02/2025 to 07/2025

- At Nuflo, I supported the development of an AI-powered menstrual health platform by contributing my software engineering skills to enhance access to menstrual products and health education across Africa.

MAMA DIGITAL WALLET PROJECT
Kampala, Uganda

Co-founder

11/2023 to 05/2025

- During my undergraduate program, I co-founded an AI-powered maternal health application to help pregnant mothers manage their finances and also monitor their health through a personalized app
- Oversaw the end-to-end product lifecycle, from concept to deployment, ensuring the application met the needs of target users effectively.
- Collaborated with cross-functional teams, including developers, designers, and healthcare experts, to deliver a user-centric AI-driven health application.
- Secured funding for the app with the MAK RISE Seeds Grant twice.

BIG DATA ANALYTICS LAB , MAKERERE UNIVERSITY

Kampala, Uganda

Undergraduate Research Assistant

10/2023 to 05/2025

- Contributed to an IoT project as a lead coordinator to deploy uniform LoRA WAN gateways across institutions in Uganda, enabling edge computing applications and sustainable energy solutions, a first for the country.
- Configured gateways to the central server and tested IoT wireless devices before deployment , transferred data from Things Stack to the Things Network for data Analysis .
- Trained sample YoLoV models to identify the best model suite for CoDA project , a project focused on identifying chest opacities using computer vision.
- Developed the lab Website
- Collaborated on the research and paper writing for 'A Deep Learning Ultrasound-Based Model for Improving Radiographic Chest Opacities Detection', with preparations to engage in the practical model-building phase.

MAKERERE UNIVERSITY

Kampala, Uganda

Class Representative of Software Engineering Class of 2021

02/2022 to 05/2025

- Coordinated the Software Engineering Class of 2021 as their leader.
- Organized and led meetings to address the academic and social concerns of the Software Engineering Class of 2021.
- Provided support and guidance to classmates on academic policies and resources, enhancing their university experience.

NTV UGANDA

Kampala, Uganda

Software Engineering Intern

06/2024 to 08/2024

- Managed networking systems and supported IT operations across departments.
- Contributed to a QR code application to enhance user experience on Uganda's leading television platform.
- Explored Machine learning applications to improve internal systems .

MAKERERE UNIVERSITY

Kampala, Uganda

Guild Representative Council

02/2023 to 02/2024

- Represented the students' body in the Debating Union Organ of the university.
- Advocated for student interests and facilitated communication between the college and the university administration at the University.
- Collaborated with the Guild Representative Council to plan and implement student-led events and initiatives.

EDUCATION

VIRTUAL TRAINEE

03/2026

Entrepreneurship Development in Higher Education (EDHE) South Africa

Developed skills in startup incubation processes, including product development, market analysis, and investor pitching within an international virtual cohort.

BACHELOR OF SCIENCE: SOFTWARE ENGINEERING

02/2026

Makerere University, Kampala, Uganda

GPA: CGPA-4.45(First class Honours) same as (3.56/4.0)

A'LEVEL: PHYSICAL SCIENCES (PEM/ICT)

11/2020

Blessed Sacrament Kimaanya, Masaka

- Comprehensive understanding of fundamental concepts in Physics, Economics, and Mathematics essential for higher education.
- GPA: 18/20points

O'LEVEL

02/2018

Blessed Sacrament Kimaanya, Masaka

GPA: 18 aggregates

WEBSITES

- <https://github.com/bidin485>
- <https://www.linkedin.com/in/nammiro-zaharah-bidin-349357254/>
- <https://nammirozaharahbidin.netlify.app/>

CERTIFICATES

- Huawei Seeds for the Future Program (China), 06/01/24
- Certificate of appreciation Guild Representative Council, 03/01/24
- Certificate of appreciation Class Representative (Makerere University) 2023
- Certificate of Completion Julius Nyerere Leadership Training, 07/01/24
- Certificate of Completion Emerging Leaders Program Training, 01/01/25
- Certificate of Completion Peace and Security Training by Konrad Adenauer Stiftung, 11/01/25
- Certificate of completion 5G, Digital Power, AI, and Cloud Computing (Huawei Tech for Good Program)11/01/23
- Certificate of Participation Bridge for Billions and LaunchLab (Nuflo Team member), 06/01/25
- Certificate of Participation (ISACA student Summit) 11/2025
- AI Governance Certification (Securiti) 11/2025
- Certificate of completion for Deep Agents with LangGraph (Langchain) 9/2025
- 3rd Runners up Women in FinTech Hackathon 08/2023
- Makerere Engineering Society Innovation Incubation program (Outstanding participation), 04/01/25

ACHIEVEMENTS

- **IoT Gateway Project (2024):** Key contributor to deploying and configuring Uganda's first uniform IoT gateway infrastructure for edge computing and sustainable energy solutions.
- **Huawei Tech for Good Program (2023):** Practical skills in 5G, Digital Power, AI, and Cloud Computing, and came up with a poultry care IoT system to assist Uganda's poultry farmers.
- **Women in FinTech Hackathon (2023):** Top 3 finalist with a fintech app addressing financial literacy and tailored savings for campus students.
- **Nuflo (2025):** Joined Nuflo as a team member; NuFlo is a social impact initiative currently in South Africa, using a digital Pad Locator App and community-based Pad Hotspots to provide free, dignified access to menstrual products, now entering phase one and aiming to expand across Africa while leveraging modern technologies like AI to offer insights and education on menstrual health.
- **Fitspa Women Competition:** Top finalist with FinHub, a fintech app solving tuition acquisition challenges.
- **Start Hub Entrepreneurial Hackathon:** Designed a simulated self-recharging hyper-generator project as part of a team of three in 2022.
- **Tika 3D Design and Printing Competition:** Designed a 3D solution to educate girls in Uganda about their reproductive systems and the effects of contraceptive use; served as team leader.
- **Uganda Oxford IFRAM Competition (2023):** Developed an AI platform for youth to assist in employment acquisition and entrepreneurship in Northern Uganda as part of a team of three.
- **Huawei Seeds for the Future (2023):** Alumni member for a project that used IoT and embedded systems to assist poultry farmers; awarded a digital trip to China.
- **Makerere Seeds Grant (2024):** Co-founder and leading the development of a maternal health fintech solution (Mama Digital Wallet), an AI-driven project supported by the MAK RISE Seeds Grant, now at phase 1.
- **A Technical leader in a start up for successful AI application** in Uganda with over 700 users .
(<https://play.google.com/store/search?q=coursemate&c=apps&hl=en>)

REFERENCES

- Ntanda Moses, Makerere University, Kampala, +256752533355, kmntanda@gmail.com
- Ggaliwango Marvin, Makerere University, Kampala, +256754655524, mggaliwango@gmail.com
- Kanyunyuzi CleoPatra, Club Tangaza, Kampala, +256702679012, clubtangaza@gmail.com

PERSONAL INFORMATION

- Date of Birth: 08/08/2002
- Gender: Female
- Nationality: Ugandan